

STAT 516 - Homework 5

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(8 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} kx^{-2}e^{-y/2}, & x > 1, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of k .

2. **(8 points)** Let X and Y be discrete random variables with joint probability function

$$f(x, y) = \begin{cases} \frac{1}{4}, & x = 1, 2 \text{ and } y = 1, 2, \\ 0, & \text{otherwise.} \end{cases}$$

Let $U = X + Y$. Find the probability function of U .

3. **(8 points)** A policy pays 200 dollars for each lost workday and 100 dollars for each doctor visit. Let X be the number of lost workdays and Y be the number of doctor visits. The joint probability function of X and Y is given by

$X \backslash Y$	0	1	2
0	0.20	0.10	0.05
1	0.08	0.12	0.07
2	0.06	0.10	0.08
3	0.04	0.06	0.04

Calculate the probability that the policy pays at least 500 dollars.

4. **(10 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{1}{4}xy, & 0 < x < 4, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $P\left(\frac{X}{4} < Y < \frac{X}{2}\right)$. (Hint: You may use $P(Y < \frac{X}{2}) - P(Y < \frac{X}{4})$)

5. **(9 points)** Let X be the portion of an insurance claim representing damage to a building, and let Y be the portion representing damage to the remaining property. The joint density function is

$$f(x, y) = \begin{cases} 12(1 - x - y)^2, & x > 0, y > 0, x + y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the probability that the portion of the claim representing damage to the building is less than 0.25.

6. **(9 points)** Let X and Y be random losses with joint density function

$$f(x, y) = 4e^{-2x-2y}, \quad x > 0, y > 0.$$

An insurance policy reimburses the amount $X + Y$. Calculate the probability that the reimbursement is less than 2.

7. **(10 points)** A device stops running when the first of two components fails. The lifetimes of the two components, in hours, have joint density function

$$f(x, y) = \frac{x + 2y}{12}, \quad 0 < x < 2, 0 < y < 2.$$

Calculate the probability that the device fails during its first hour of operation, i.e., $P(\min(X, Y) < 1)$. (Hint: use the complement rule)

8. **(10 points)** Let X and Y have joint density function

$$f(x, y) = \begin{cases} \frac{1}{6}, & 0 < x < 3, x < y < x + 2, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal density function of Y .

9. **(9 points)** Let X and Y be discrete random variables with joint probability function

$$f(x, y) = \begin{cases} \frac{x + y}{21}, & (x, y) = (1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2), \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal probability function of X .

10. **(10 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{2}{5}(x + y), & 0 < x < 1, x < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal density function of Y .

11. **(9 points)** Let X and Y have a uniform joint distribution over the region

$$\{(x, y) : x^2 + y^2 \leq 4, x > 0, y > 0\}.$$

Find the marginal density function of X .

1. (8 points) Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} kx^{-2}e^{-y/2}, & x > 1, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of k .

Joint Density total probability must = 1

$$\int_1^{\infty} \int_0^{\infty} kx^{-2}e^{-y/2} dy dx = 1 \xrightarrow{\text{separates into x-part \& y-part}} k \left(\int_1^{\infty} x^{-2} dx \right) \left(\int_0^{\infty} e^{-y/2} dy \right) = 1$$

$$\int_1^{\infty} x^{-2} dx = \left[-x^{-1} \right]_1^{\infty} = 0 - (-1) = 1$$

$$\int_0^{\infty} e^{-y/2} dy = \left[-2e^{-y/2} \right]_0^{\infty} = 0 - (-2) = 2$$

$$k(1)(2) = 1$$

$$2k = 1$$

$$k = 1/2$$

2. (8 points) Let X and Y be discrete random variables with joint probability function

$$f(x, y) = \begin{cases} \frac{1}{4}, & x = 1, 2 \text{ and } y = 1, 2, \\ 0, & \text{otherwise.} \end{cases} \rightarrow \text{All Possible Pairs:}$$

$(X, Y) = (1, 1) (2, 1)$
 $(1, 2) (2, 2)$

Each pair with prob. = $1/4$

Let $U = X + Y$. Find the probability function of U .

$$\begin{array}{l} (1, 1) \rightarrow U = 1 + 1 = 2 \\ (1, 2) \rightarrow U = 1 + 2 = 3 \\ (2, 1) \rightarrow U = 2 + 1 = 3 \\ (2, 2) \rightarrow U = 2 + 2 = 4 \end{array} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{Possible values of } U: 2, 3, 4 \\ 2, \text{ one pair with prob} = 1/4 \\ 3, \text{ two pairs with prob} = 1/2 \\ 4, \text{ one pair with prob} = 1/4 \end{array}$$

$$p_U(x) = \begin{cases} 1/4, & U = 2 \\ 1/2, & U = 3 \\ 1/4, & U = 4 \\ 0, & \text{otherwise} \end{cases}$$

$$p_U(2) = 1/4, p_U(3) = 1/2, p_U(4) = 1/4$$

3. (8 points) A policy pays 200 dollars for each lost workday and 100 dollars for each doctor visit. Let X be the number of lost workdays and Y be the number of doctor visits. The joint probability function of X and Y is given by

		Y		
$X \backslash Y$		0	1	2
X	0	0.20	0.10	0.05
	1	0.08	0.12	0.07
	2	0.06	0.10	0.08
	3	0.04	0.06	0.04

Calculate the probability that the policy pays at least 500 dollars. $P(200X + 100Y \geq 500)$
 $\div 100 \quad \div 100$

$$\text{Total payment} = 200X + 100Y$$

$$P(2X + Y \geq 5)$$

Case 1: $X=0$ $\rightarrow Y \geq 5$ none, $Y=0,1,2$ are all < 5 .	Case 2: $X=1$ $\rightarrow 2+Y \geq 5$ none, $Y=0,1,2$ are all < 5 .	Case 3: $X=2$ $\rightarrow 4+Y \geq 5$ $\bullet Y=1,2 \checkmark$	Case 4: $X=3$ $\rightarrow 6+Y \geq 5$ $\bullet Y=0,1,2 \checkmark$
		$\left. \begin{matrix} (2,1) \\ (2,2) \end{matrix} \right\}$	$\left. \begin{matrix} (3,0) \\ (3,1) \\ (3,2) \end{matrix} \right\}$

$$P(2X + Y \geq 5) = 0.10 + 0.08 + 0.04 + 0.06 + 0.04 = \underline{0.32}$$

$$P(200X + 100Y \geq 500) = 0.32$$

4. (10 points) Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{1}{4}xy, & 0 < x < 4, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $P\left(\frac{x}{4} < Y < \frac{x}{2}\right)$. (Hint: You may use $P(Y < \frac{x}{2}) - P(Y < \frac{x}{4})$) for $2 < X < 4$

upper bound: $y=1$

$\bullet Y$ cannot go above 1, so $X=2$ and $X=4$

$$\frac{x}{4} < y < 1, 0 < x < 2$$

$$P\left(\frac{x}{4} < Y < \frac{x}{2}\right) = \int_0^2 \int_{x/4}^{x/2} \frac{1}{4}xy \, dy \, dx + \int_2^4 \int_{x/4}^1 \frac{1}{4}xy \, dy \, dx = \frac{3}{32} + \frac{9}{32} = \frac{12}{32} = \frac{3}{8}$$

$$\int_0^2 \int_{x/4}^{x/2} \frac{1}{4}xy \, dy \, dx = \int_0^2 \left[\frac{1}{4}x \left[\frac{y^2}{2} \right]_{x/4}^{x/2} \right] dx$$

$$= \int_0^2 \frac{x}{8} \left[\left(\frac{x}{2} \right)^2 - \left(\frac{x}{4} \right)^2 \right] dx = \int_0^2 \frac{x}{8} \left(\frac{x^2}{4} - \frac{x^2}{16} \right) dx = \int_0^2 \frac{x}{8} \left(\frac{3x^2}{16} \right) dx$$

$$= \int_0^2 \frac{3x^3}{128} dx = \frac{3}{128} \left[\frac{x^4}{4} \right]_0^2 = \frac{3}{128} \cdot 4 = \frac{12}{128} = \frac{3}{32}$$

$$\int_2^4 \int_{x/4}^1 \frac{1}{4}xy \, dy \, dx = \int_2^4 \left[\frac{1}{4}x \left[\frac{y^2}{2} \right]_{x/4}^1 \right] dx$$

$$= \int_2^4 \frac{x}{8} \left[1^2 - \left(\frac{x}{4} \right)^2 \right] dx = \int_2^4 \frac{x}{8} \left(1 - \frac{x^2}{16} \right) dx = \int_2^4 \left(\frac{x}{8} - \frac{x^3}{128} \right) dx$$

$$= \left[\frac{x^2}{16} - \frac{x^4}{512} \right]_2^4 = \frac{1}{2} - \frac{7}{32} = \frac{16}{32} - \frac{7}{32} = \frac{9}{32}$$

$$P\left(\frac{x}{4} < Y < \frac{x}{2}\right) = \frac{3}{8}$$

5. (9 points) Let X be the portion of an insurance claim representing damage to a building, and let Y be the portion representing damage to the remaining property. The joint density function is

$$f(x, y) = \begin{cases} 12(1-x-y)^2, & x > 0, y > 0, x+y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$y < 1-x$
 $\rightarrow 0 < y < 1-x$

Calculate the probability that the portion of the claim representing damage to the building is less than 0.25. $P(X < 0.25) \rightarrow 0 < x < 0.25$

$$\begin{aligned} P(X < 0.25) &= \int_0^{0.25} \int_0^{1-x} 12(1-x-y)^2 dy dx \\ &\quad \begin{matrix} u = 1-x-y \\ du = -dy \end{matrix} \\ &= \int_0^{0.25} \int_{1-x}^0 12u^2(-du) dx = \int_0^{0.25} \int_0^{1-x} 12u^2 du dx \\ &= \int_0^{0.25} 12 \left[\frac{u^3}{3} \right]_0^{1-x} dx = \int_0^{0.25} 4(1-x)^3 dx = \left[-(1-x)^4 \right]_0^{0.25} \\ &= -(1-0.25)^4 - [- (1-0)^4] = -(0.75)^4 + 1 = 1 - \left(\frac{3}{4}\right)^4 = 1 - \frac{81}{256} \end{aligned}$$

$$P(X < 0.25) = \frac{175}{256}, \text{ or } 0.684$$

6. (9 points) Let X and Y be random losses with joint density function

$$f(x, y) = 4e^{-2x-2y}, \quad x > 0, y > 0, x+y < 2$$

$y < 2-x$
 $\rightarrow 0 < y < 2-x$

An insurance policy reimburses the amount $X + Y$. Calculate the probability that the reimbursement is less than 2. $P(X+Y < 2)$

$$\begin{aligned} P(X+Y < 2) &= \int_0^2 \int_0^{2-x} 4e^{-2x-2y} dy dx = 4 \int_0^2 e^{-2x} \left[\int_0^{2-x} e^{-2y} dy \right] dx \\ &= 4 \int_0^2 e^{-2x} \left[-\frac{1}{2} e^{-2y} \right]_0^{2-x} dx \\ &= 2 \int_0^2 e^{-2x} (1 - e^{-2(2-x)}) dx = 2 \int_0^2 e^{-2x} dx - 2 \int_0^2 e^{-4} dx = (1 - e^{-4}) - 4e^{-4} \end{aligned}$$

$$P(X+Y < 2) = 1 - 5e^{-4}, \text{ or } 0.908$$

7. (10 points) A device stops running when the first of two components fails. The lifetimes of the two components, in hours, have joint density function

$$f(x, y) = \frac{x+2y}{12}, \quad 0 < x < 2, \quad 0 < y < 2.$$

Calculate the probability that the device fails during its first hour of operation, i.e., $P(\min(X, Y) < 1)$.

(Hint: use the complement rule) $\rightarrow X \geq 1, Y \geq 1$, for $1 < X < 2, 1 < Y < 2$

$$P(\min(X, Y) \geq 1) = \int_1^2 \int_1^2 \frac{x+2y}{12} dy dx$$

$$= \frac{1}{12} \int_1^2 \int_1^2 (x+2y) dy dx = \frac{1}{12} \int_1^2 [xy + y^2]_1^2 dx = \frac{1}{12} \int_1^2 (x+3) dx$$

$$= \frac{1}{12} \left[\frac{x^2}{2} + 3x \right]_1^2 = \frac{1}{12} \left(\frac{4}{2} + 6 - \frac{1}{2} - 3 \right) = \frac{3}{8}$$

$$\begin{aligned} y=2 &: 2x+4 \\ y=1 &: x+1 \\ (2x+4) - (x+1) &= x+3 \end{aligned}$$

$$\begin{aligned} x=2 &: \frac{4}{2} + 6 = 8 \\ x=1 &: \frac{1}{2} + 3 = \frac{7}{2} \\ 8 - \frac{7}{2} &= \frac{9}{2} \end{aligned}$$

$$P(\min(X, Y) < 1) = 1 - \frac{3}{8}$$

$$P(\min(X, Y) < 1) = \frac{5}{8}$$

8. (10 points) Let X and Y have joint density function

$$f(x, y) = \begin{cases} \frac{1}{6}, & 0 < x < 3, \quad x < y < x+2, \\ 0, & \text{otherwise.} \end{cases}$$

$$\begin{aligned} x < y < x+2 \\ -2 \quad -2 \quad -2 \\ x-2 < y-2 < x \end{aligned}$$

Fixed y :
 $\rightarrow \max(0, y-2) < x < \min(3, y)$

Determine the marginal density function of Y . $f_Y(y) = \int f(x, y) dx$

$$0 < x < 3 \quad \text{and} \quad x < y < x+2$$

$$\left. \begin{aligned} x = \min(0, y-2) \\ x = \max(3, y) \end{aligned} \right\} 0 < y < 5 \left\{ \begin{aligned} 0 < y < 2 \\ 2 \leq y < 3 \\ 3 \leq y < 5 \end{aligned} \right.$$

$$\text{case 1: } 0 < y < 2$$

$$\rightarrow \int_0^y \frac{1}{6} dx = y/6$$

$$0 < x < y$$

$$\text{case 2: } 2 \leq y < 3$$

$$\rightarrow \int_{y-2}^y \frac{1}{6} dx = \frac{1}{3}$$

$$y-2 < x < y$$

$$\text{case 2: } 3 \leq y < 5$$

$$\rightarrow \int_{y-2}^3 \frac{1}{6} dx = \frac{5-y}{6}$$

$$y-2 < x < 3$$

Answer Below:

$$f_Y(y) = \begin{cases} \int_0^y \frac{1}{6} dx, & 0 < y < 2 \\ \int_{y-2}^y \frac{1}{6} dx, & 2 \leq y < 3 \\ \int_{y-2}^3 \frac{1}{6} dx, & 3 \leq y < 5 \\ 0, & \text{otherwise} \end{cases}, \text{ or } f_Y(y) = \begin{cases} y/6, & 0 < y < 2 \\ 1/3, & 2 \leq y < 3 \\ \frac{5-y}{6}, & 3 \leq y < 5 \\ 0, & \text{otherwise} \end{cases}$$

9. (9 points) Let X and Y be discrete random variables with joint probability function

$$f(x, y) = \begin{cases} \frac{x+y}{21}, & (x, y) = (1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2), \quad \begin{matrix} x=1,2,3 \\ y=1,2 \end{matrix} \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal probability function of X . $p_X(x) = \sum_y f(x, y)$

$$p_X(x) = f(x, 1) + f(x, 2)$$

$$x=1:$$

$$p_X(1) = f(1, 1) + f(1, 2) = \frac{1+1}{21} + \frac{1+2}{21}$$

$$p_X(1) = 5/21$$

$$x=2:$$

$$p_X(2) = f(2, 1) + f(2, 2) = \frac{2+1}{21} + \frac{2+2}{21}$$

$$p_X(2) = 7/21$$

$$x=3:$$

$$p_X(3) = f(3, 1) + f(3, 2) = \frac{3+1}{21} + \frac{3+2}{21}$$

$$p_X(3) = 9/21$$

$$p_X(x) = \begin{cases} 5/21, & x=1 \\ 7/21, & x=2 \\ 9/21, & x=3 \\ 0, & \text{otherwise} \end{cases}$$

$$pdf = 1 = 5/21 + 7/21 + 9/21 = 1 \quad \checkmark$$

10. (10 points) Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{2}{5}(x + y), & 0 < x < 1, x < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Fix y : $0 < x < \min(1, y)$

$0 < y < 1$, $0 < x < y$

$1 \leq y < 2$, $0 < x < 1$

Find the marginal density function of Y . $f_Y(y) = \int f(x, y) dx$

Case 1:

$\rightarrow 0 < y < 1$

$$f_Y(y) = \int_0^y \frac{2}{5}(x+y) dx = \frac{2}{5} \left[\frac{x^2}{2} + yx \right]_0^y = \frac{2}{5} \left(\frac{y^2}{2} + y^2 \right) = \frac{2}{5} \left(\frac{3y^2}{2} \right) = \frac{3y^2}{5}$$

$0 < x < y$

Case 2:

$\rightarrow 1 \leq y < 2$

$$f_Y(y) = \int_0^1 \frac{2}{5}(x+y) dx = \frac{2}{5} \left[\frac{x^2}{2} + yx \right]_0^1 = \frac{2}{5} \left(\frac{1}{2} + y \right) = \frac{1+2y}{5}$$

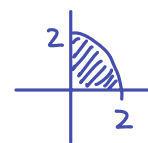
$0 < x < 1$

$$f_Y(y) = \begin{cases} \int_0^y \frac{2}{5}(x+y) dx, & 0 < y < 1 \\ \int_0^1 \frac{2}{5}(x+y) dx, & 1 \leq y < 2 \\ 0, & \text{otherwise} \end{cases}, \text{ or } f_Y(y) = \begin{cases} 3y^2/5, & 0 < y < 1 \\ (1+2y)/5, & 1 \leq y < 2 \\ 0, & \text{otherwise} \end{cases}$$

11. (9 points) Let X and Y have a uniform joint distribution over the region

$$\{(x, y) : x^2 + y^2 \leq 4, x > 0, y > 0\}.$$

Find the marginal density function of X .



$$\bullet 0 < x < 2$$

$$y \leq \sqrt{4-x^2} :$$

$$\bullet 0 < y < \sqrt{4-x^2}$$

radius = 2

Area of quarter circle :

$$\frac{1}{4}(\pi r^2) = \frac{1}{4}\pi 2^2 = \pi$$

$$f(x, y) = C$$

$$\iint f(x, y) dA = 1 \rightarrow C \underbrace{\iint dA}_{\text{Area}} = 1 \rightarrow C(\text{Area}) = 1$$

$$C = \frac{1}{\text{Area}}$$

$$f(x, y) = \frac{1}{\text{Area}} = \frac{1}{\pi}$$

$$f_X(x) = \int_0^{\sqrt{4-x^2}} \frac{1}{\pi} dy = \frac{1}{\pi} [y]_0^{\sqrt{4-x^2}} = \frac{1}{\pi} \sqrt{4-x^2}$$

$$f_X(x) = \begin{cases} \frac{1}{\pi} \sqrt{4-x^2}, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$